

Project Overview: Bias-Variance Trade-off & Auto Dataset Analysis

Question 1 – Flexibility vs. Inflexibility in Learning Methods

Analyzed four scenarios to determine when flexible or inflexible statistical learning methods are preferable:

- **(a) Large n , small p :**
Flexible methods perform well due to sufficient data to model complex patterns without overfitting.
- **(b) Small n , large p :**
Inflexible models generalize better; flexible models tend to overfit due to high dimensionality and limited observations.
- **(c) Highly non-linear relationships:**
Flexible models adapt better to non-linearity, reducing underfitting.
- **(d) High error variance:**
Inflexible models help avoid fitting noise, improving generalization.

Question 2 – Bias-Variance Decomposition and Visualization

(a) Simulation of Error Components

Simulated five curves over increasing model flexibility using R:

- Bias^2 ($\downarrow$), Variance ($\uparrow$)
- Training Error ($\downarrow$)
- Test Error (U-shaped)
- Bayes Error (Constant)

Visualized using ggplot2, capturing key trade-offs in model complexity.

(b) Explanation of Curve Shapes

- **Bias²:** High in simple models, drops with flexibility.
- **Variance:** Low in simple models, increases as models overfit.
- **Training Error:** Decreases rapidly with flexibility.
- **Test Error:** Initially falls, then rises due to overfitting.
- **Bayes Error:** Flat, representing irreducible noise.

Question 3 – Auto Dataset Analysis

(a) Predictor Types

Identified:

- **Quantitative:** mpg, cylinders, displacement, horsepower, weight, acceleration, year, origin
- **Qualitative:** name

(b) Ranges of Quantitative Predictors

Used `range()` to determine min/max for each quantitative feature.

(c) Central Tendency & Dispersion

Computed mean and standard deviation for each quantitative predictor using `summary()`.

(d) Subset Analysis

Removed rows 10–85; recomputed range, mean, and SD for comparison. Changes were minor but measurable in some predictors (e.g., mpg, weight).

(e) Graphical Exploration

Used `ggpairs()` to create a scatterplot matrix for quantitative variables. Strong correlations found:

- mpg vs weight, displacement, horsepower: strong negative
- mpg vs year: positive

(f) Predictive Indicators for mpg

Scatterplots showed strong predictive relationships between mpg and:

- **Negative:** weight, horsepower, displacement, cylinders
- **Positive:** year, acceleration (moderate)

These findings support their inclusion in regression modeling for fuel efficiency.

Tools and Environment

- **Language:** R
- **Libraries:** ggplot2, reshape2, GGally
- **Data:** Cleaned Auto.data (missing values removed)